Matemàtiques B
4t ESO

SOLUCIONS DE LA FITXA · UNITAT 1

Nombres reals

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Pots comprovar els d'intervals, valor absolut, radicals, racionalització i logaritmes amb les eines interactives dels apunts de la unitat.

1 Classificació dels nombres

1. ★☆☆

a) $\boxed{\mathbb{Q}}$

b) $7{,}\widehat{9} = \frac{79-7}{9} = 8$: $\boxed{\mathbb{N}}$

c) No és un nombre real.

d) $\boxed{\mathbb{I}}$ (infinites xifres sense període)

e) -2: $\boxed{\mathbb{Z}}$

f) 4: $\boxed{\mathbb{N}}$

g) $\boxed{\mathbb{I}}$ (les xifres no es repeteixen en blocs)

2 Intervals i semirectes

2. ★☆☆

a) $x \leq 2$



b) $1 \leq x \leq 5$



c) $2 < x \leq 4$



d) $x > -7$

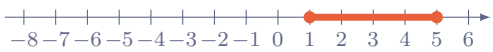


3. ★★☆

a) $\boxed{(-\infty, 5]}$



b) $\boxed{[1, 5]}$



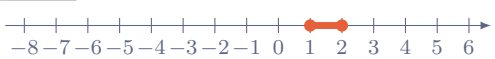
c) $\boxed{\mathbb{R}}$



d) $\boxed{(-\infty, 5]}$



e) $\boxed{[1, 2]}$



f) $\boxed{[1, 5]}$



g) $\boxed{\varnothing}$ (el 2 és de A però no de C)

h) $\boxed{\varnothing}$

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3 Valor absolut

4. ★★☆

a) $\boxed{(-5, 5)}$



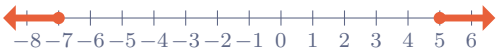
b) $\boxed{(-\infty, -4] \cup [4, +\infty)}$



c) $-9 \leq x + 2 \leq 9$: $\boxed{[-11, 7]}$



d) $x + 1 \leq -6$ o $x + 1 \geq 6$: $\boxed{(-\infty, -7] \cup [5, +\infty)}$



4 Potències

5. ★☆☆

a) $\boxed{-32}$ **b)** $\boxed{-32}$ **c)** $\boxed{1}$ **d)** $\boxed{-1}$ **e)** $\boxed{9}$

f) $\boxed{8}$ **g)** $\boxed{\frac{9}{4}}$ **h)** $\boxed{\frac{1}{16}}$ **i)** $\boxed{\frac{9}{4}}$ **j)** $\boxed{5}$

6. ★★☆

a) $\left[-\frac{2^2}{3^5}\right]^2 = \boxed{\frac{2^4}{3^{10}}}$

b) $\frac{-2^{17}}{-2^{-2}} = \boxed{2^{19}}$

c) $2^7 \cdot 3^{-4} \cdot 5 = \boxed{\frac{640}{81}}$

d) $\frac{2^4 \cdot 3^{-6}}{2^3 \cdot 3^{-5}} = \boxed{\frac{2}{3}}$

5 Radicals

7. ★☆☆

a) $\boxed{4}$ **b)** $\boxed{-2}$ **c)** $\boxed{10}$

d) $\boxed{-2}$ **e)** $\boxed{0{,}6}$ **f)** No és real.

g) $\boxed{1}$ **h)** $\boxed{-1}$ **i)** $\boxed{1}$ (i -1, perquè $(-1)^2 = 1$)

8. ★★☆

a) $\sqrt{2^8 \cdot 5} = \boxed{16\sqrt{5}}$

b) $\sqrt[3]{2^3 \cdot 3^3 \cdot 5} = \boxed{6\sqrt[3]{5}}$

c) $\sqrt[3]{2^6 \cdot 3^4} = \boxed{12\sqrt[3]{3}}$

d) $\sqrt[3]{2^{10}} = \boxed{8\sqrt[3]{2}}$

e) $\boxed{2^2 \cdot 3^3 \cdot 5^6 \sqrt[4]{2 \cdot 3^3 \cdot 5^3}}$

f) $\boxed{3^4 \cdot 5^5 \cdot 7^2 \sqrt{5 \cdot 7}}$

g) $\boxed{a^3 b^3 c^4 \sqrt{abc}}$

h) $\boxed{x^5 y^6 z^9 \sqrt[5]{x^2 z^3}}$

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9. ★★☆

a) $\boxed{\sqrt{\frac{18}{25}}}$ **b)** $\boxed{\sqrt[3]{40}}$ **c)** $\boxed{\sqrt[3]{56}}$ **d)** $\boxed{\sqrt[4]{5120}}$

e) $\boxed{\sqrt[4]{\frac{3}{8}}}$ **f)** $\boxed{\sqrt[3]{25}}$ **g)** $\boxed{\sqrt[5]{3645}}$ **h)** $\boxed{\sqrt[4]{\frac{1}{32}}}$

10. ★☆☆

a) $\boxed{2}$ (oposades: $\pm\sqrt{16} = \pm 4$) **b)** $\boxed{0}$ ($\sqrt{-16} \notin \mathbb{R}$)

c) $\boxed{1}$, positiva **d)** $\boxed{1}$, negativa ($\sqrt[3]{-8} = -2$)

11. ★☆☆

a) $\boxed{2^{\frac{19}{12}}}$ **b)** $\boxed{2^{-\frac{1}{6}}}$ **c)** $\boxed{3^{\frac{5}{4}}}$ **d)** $\boxed{5^{\frac{3}{2}}}$

12. ★★☆

a) $\boxed{\sqrt[6]{5^5}}$ **b)** $\boxed{\sqrt[12]{2^4 \cdot 7^3}}$ **c)** $\boxed{\sqrt[6]{3^7} = 3\sqrt[6]{3}}$ **d)** $\boxed{a^{\frac{25}{12}} = a^2\,\sqrt[12]{a}}$

e) $\boxed{3^{-\frac{3}{10}} = \frac{1}{\sqrt[10]{27}}}$ **f)** $\boxed{\sqrt[6]{5}}$ **g)** $\boxed{\sqrt[8]{8}}$ **h)** $\boxed{2^{\frac{5}{6}} = \sqrt[6]{32}}$

13. ★★☆

a) $3\sqrt{5} - 5\sqrt{5} - 2\sqrt{5} = \boxed{-4\sqrt{5}}$ **b)** $5\sqrt{3} - 7\sqrt{3} + 15\sqrt{3} - 2\sqrt{3} = \boxed{11\sqrt{3}}$

c) $5\sqrt{7} + 3\sqrt{7} - 4\sqrt{7} = \boxed{4\sqrt{7}}$ **d)** $2\sqrt{5} + \sqrt{5} + 10\sqrt{5} = \boxed{13\sqrt{5}}$

e) $4\sqrt{2} - 2\sqrt{2} + 18\sqrt{2} - 20\sqrt{2} = \boxed{0}$ **f)** $7\sqrt[3]{3} - 2\sqrt[3]{3} + 9\sqrt[3]{3} - 20\sqrt[3]{3} = \boxed{-6\sqrt[3]{3}}$

6 Racionalització

14. ★★☆

a) $\boxed{\frac{2\sqrt{7}}{7}}$ **b)** $\boxed{\frac{\sqrt{3}}{3}}$ **c)** $\boxed{\sqrt{6}}$ **d)** $\boxed{3\sqrt[3]{2}}$

e) $\boxed{\frac{\sqrt[3]{100}}{4}}$ **f)** $\boxed{10 + 5\sqrt{3}}$ **g)** $\boxed{\sqrt{15} + 3}$ **h)** $\boxed{\frac{2\sqrt{2} - 1}{7}}$

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7 Logaritmes

15. ★☆☆

a) $\boxed{3}$ **b)** $\boxed{4}$ **c)** $\boxed{3}$ **d)** $\boxed{-4}$

e) $\boxed{-4}$ **f)** $\boxed{-1}$ **g)** $\boxed{\frac{1}{3}}$ **h)** $\boxed{3}$

16. ★☆☆

a) $\boxed{x = 3}$ **b)** $\boxed{x = 256}$ **c)** $\boxed{x = 5}$ **d)** $\boxed{x = 5}$

e) $\boxed{x = -4}$ **f)** $\boxed{x = 4}$ **g)** $\boxed{x = 1}$ **h)** $\boxed{x = 0}$

17. ★★☆

a) $0{,}31 + 1{,}4 = \boxed{1{,}71}$ **b)** $3 \cdot 0{,}31 + 1{,}4 = \boxed{2{,}33}$

8 Ampliació: nombres, intervals, valor absolut i potències

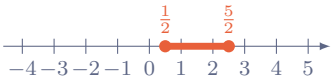
18. ★★☆

a) $\frac{25}{9} + \frac{31}{9} - \frac{10}{9} = \boxed{\frac{46}{9}}$ **b)** $\frac{17}{9} + \frac{1}{9} = \boxed{2}$

c) $\frac{1}{6} \cdot \frac{14}{9} = \boxed{\frac{7}{27}}$ **d)** $\frac{37}{30} - \frac{5}{11} = \boxed{\frac{257}{330}}$

19. ★★☆

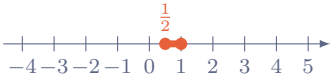
a) $\boxed{[\frac{1}{2}, \frac{5}{2}]}$



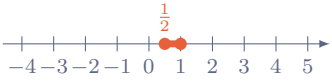
b) $\boxed{(-\infty, \frac{5}{2}]}$



c) $\boxed{[\frac{1}{2}, 1]}$



d) $\boxed{[\frac{1}{2}, 1]}$



e) $\boxed{(-\infty, 4)}$



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20. ★★☆

a) $\boxed{(-\infty,\,-1)\cup(2,\,+\infty)}$



b) $\boxed{[-5,1]}$



c) $\boxed{(-\infty,3]\cup[7,\,+\infty)}$



d) $\boxed{\varnothing}$ (un valor absolut no pot ser negatiu)

21. ★★★

a) $\boxed{|x-4|<3}$ **b)** $\boxed{|x+2|\leq 3}$ **c)** $\boxed{|x-2|>3}$

22. ★★☆ $\boxed{|T-4|\leq 1{,}5}$, és a dir, $T\in[2{,}5;\ 5{,}5]$. Una lectura de $5{,}7\,^\circ$C $\boxed{\text{no és acceptable}}$: $|5{,}7-4|=1{,}7>1{,}5$.

23. ★★★ $\dfrac{2^{-2}3^{-2}\cdot 3^6\cdot 2^5}{2^4\cdot 3^3\cdot 3^{-1}}=2^{-1}\cdot 3^2=\boxed{\dfrac{9}{2}}$; $\dfrac{4\cdot\frac{1}{3}}{8}=\boxed{\dfrac{1}{6}}$; $2^2=\boxed{4}$; $3^3=\boxed{27}$.

9 Ampliació: radicals i racionalització

24. ★★☆

a) $\boxed{\sqrt{5}}$ **b)** $\boxed{2\sqrt{3}}$ **c)** $\boxed{\dfrac{\sqrt{10}}{2}}$ **d)** $\boxed{\dfrac{3\sqrt{5}}{5}}$

e) $\boxed{\dfrac{\sqrt[3]{12}}{2}}$ **f)** $\boxed{\dfrac{3\sqrt[3]{2}}{4}}$ **g)** $\boxed{\dfrac{3\sqrt{y}}{x}}$ **h)** $\boxed{\dfrac{\sqrt{xy}}{y}}$

i) $\boxed{\dfrac{4}{5}}$ **j)** $\boxed{\sqrt{3}}$ **k)** $\boxed{3y\sqrt[3]{xy^2}}$ **l)** $\boxed{6\sqrt[3]{3-y}}$

m) $\boxed{-\dfrac{2}{3}\sqrt[4]{27}\,\sqrt[3]{4}}$

25. ★★★

a) $\boxed{\sqrt{3}-\sqrt{2}}$ **b)** $\boxed{2\sqrt{2}+2}$ **c)** $\boxed{\dfrac{3\sqrt{2}+2}{7}}$

d) $\boxed{2\sqrt{2}-3}$ **e)** $\boxed{\dfrac{\sqrt{5}-4}{11}}$ **f)** $\boxed{\dfrac{\sqrt{x}+\sqrt{y}}{x-y}}$

g) $\boxed{\dfrac{2\sqrt{y}+y\sqrt{2}}{2-y}}$ **h)** $\boxed{\dfrac{2x\sqrt{y}+3y\sqrt{x}}{4x-9y}}$ **i)** $\boxed{\dfrac{5(\sqrt{3}+\sqrt{2})}{2}}$

j) $\boxed{\dfrac{4-\sqrt{2}}{7}}$ **k)** $\boxed{3+2\sqrt{2}}$

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26. ★★☆

a) $\boxed{14\sqrt{2}}$ **b)** $\boxed{0}$ **c)** $\boxed{7\sqrt{3}}$ **d)** $\boxed{0}$

e) $\boxed{6\sqrt{2}}$ **f)** $\boxed{a\sqrt{3}}$ **g)** $\boxed{(2a-2)\sqrt{5}}$ **h)** $\boxed{22\sqrt{6}-12\sqrt{2}}$

i) $\boxed{\dfrac{10\sqrt{3}}{3}}$ **j)** $\boxed{\dfrac{11\sqrt{2}}{2}}$ **k)** $\boxed{-\dfrac{13\sqrt{3}}{2}}$ **l)** $\boxed{\sqrt{2}+\sqrt{3}}$

m) $\boxed{\sqrt{3}}$ **n)** $\boxed{5\sqrt{5}}$

(Als apartats f) i g) suposem $a \geq 0$.)

27. ★★★

a) $\boxed{5+2\sqrt{6}}$ **b)** $\boxed{19}$

c) $\boxed{-2}$ **d)** $\boxed{6}$

e) $\dfrac{(\sqrt{3}+1)^2-(\sqrt{3}-1)^2}{2} = \boxed{2\sqrt{3}}$ **f)** $\dfrac{\sqrt{2}}{2}+\sqrt{2}-1 = \boxed{\dfrac{3\sqrt{2}}{2}-1}$

28. ★★☆ Costat: $\sqrt{50} = \boxed{5\sqrt{2}\text{ cm}}$. Diagonal: $\sqrt{2}\cdot 5\sqrt{2} = \boxed{10\text{ cm}}$.

10 Ampliació: logaritmes i problemes

29. ★★☆

a) $\boxed{x=7}$ **b)** $\boxed{x=8}$ **c)** $\boxed{x=2}$ **d)** $\boxed{x=4}$

30. ★★☆

a) $2\log 2+\log 3 = \boxed{1{,}079}$ **b)** $\log\frac{10}{2} = 1-\log 2 = \boxed{0{,}699}$

c) $\log 3-\log 2 = \boxed{0{,}176}$ **d)** $\frac{1}{2}(\log 2+2\log 3) = \boxed{0{,}6275}$

e) $\log\frac{6}{100} = \log 2+\log 3-2 = \boxed{-1{,}222}$

31. ★★☆

a) $\boxed{2+3\log x-\log y}$ **b)** $\boxed{\frac{1}{2}(\log x-\log y)}$

c) $\boxed{E=\dfrac{3x^2}{y}}$ **d)** $\boxed{E=\dfrac{100}{x}}$

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32. ★☆☆

a) $\boxed{2{,}727}$ **b)** $\boxed{-5{,}907}$ **c)** $\boxed{6{,}644}$ **d)** $\boxed{1{,}946}$

33. ★★☆ $100 \cdot 2^t = 10^6 \Rightarrow 2^t = 10^4 \Rightarrow t = \log_2 10^4 = \dfrac{4}{\log 2} \approx \boxed{13{,}29 \text{ hores}}$.

34. ★★☆ $1500 \cdot 1{,}04^t = 2000 \Rightarrow 1{,}04^t = \frac{4}{3} \Rightarrow t = \dfrac{\log \frac{4}{3}}{\log 1{,}04} \approx 7{,}33$. Calen $\boxed{8 \text{ anys}}$.

35. ★★★ pH $= \boxed{3}$; pH $= -\log(2{,}5 \cdot 10^{-5}) \approx \boxed{4{,}6}$. És més àcida $\boxed{\text{la primera}}$.

36. ★★★ $2^3 = 8 < 10 \Rightarrow \log_2 10 > 3$; $3^2 = 9 < 10 < 27 = 3^3 \Rightarrow 2 < \log_3 10 < 3$. Per tant, $\boxed{\log_3 10 < 3 < \log_2 10}$.

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